



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. (General Degree)

Second Year Semester I Examination – October / November 2014

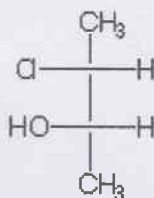
CHE 2202 – ORGANIC CHEMISTRY II

Answer any four questions.

Time: 02 hours

1. (a). Name the following compounds using RS nomenclature. Draw necessary steps and write IUPAC names of the compounds.

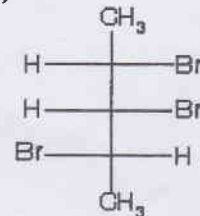
i).



ii).



iii).



(12 marks)

- b). Draw the Fischer projection of (2R,4R)-2,4-dibromopentane

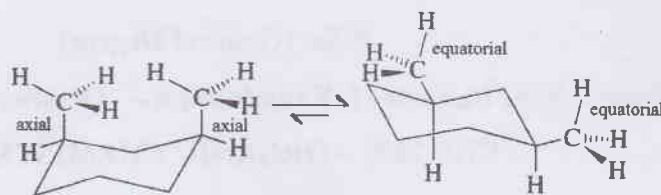
(04 marks)

- c). Draw the conformations of Butane using Newman projection formulae and plot the potential energy vs angle of rotation curve for the rotation of C2 – C3 bond through 360°C

(09 marks)

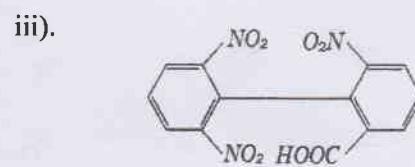
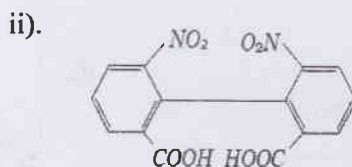
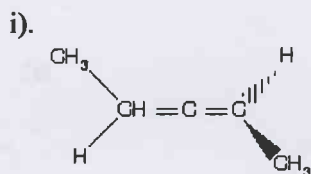
2.

- (a). Chair conformations of *cis* 1,3-Dimethylcyclohexane show two different energy levels, while *trans* 1,3-Dimethylcyclohexane show same energy levels. Draw necessary diagrams and discuss this observation.

*cis* 1,3-Dimethylcyclohexane

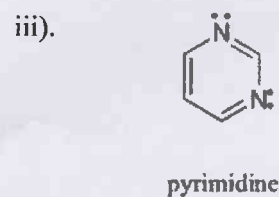
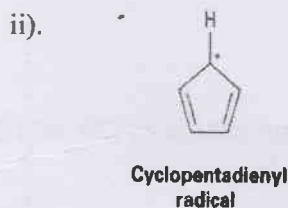
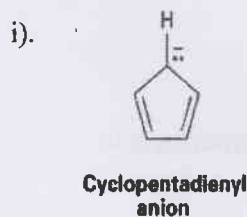
(06 marks)

- (b). Discuss the optical activity of the following compounds



(06 marks)

- (c). Explain the aromaticity of the following compounds using Huckels rule



(06 marks)

- (d). i. List out the limitations of Friedel-Craft alkylation reactions (03 marks)

- ii. Outline the reactions of synthesis of Benzoic acid from Benzene via *n* - propyl benzene

(04 marks)

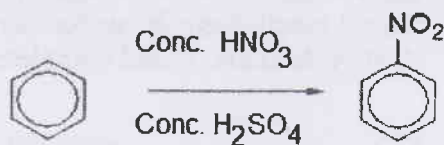
3.

- (a). Benzenesulfonic acid is prepared from the sulfonation of Benzene using concentrated Sulfuric acid. Write the mechanisms of this reaction.



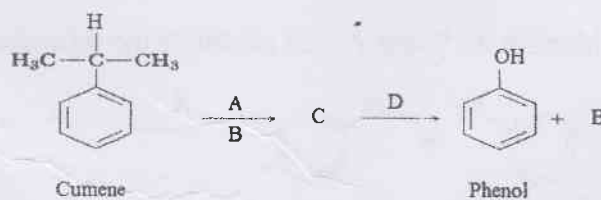
(07 marks)

- (b). The nitration reaction of Benzene can be expressed as follows. Write the detailed mechanism of the reaction and discuss the energy diagram of the reaction.



(09 marks)

- (c). Synthesis of Phenol using Cumene is given below. Identify A, B, C and D, and complete the following reactions.



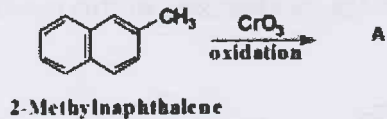
(04 marks)

- (d). Elaborate a method to synthesis of p-methyl phenol in the laboratory.

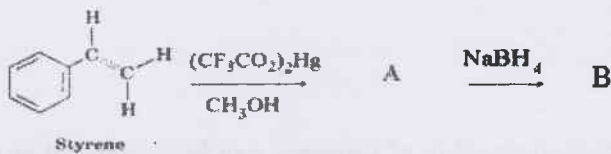
(05 marks)

4.

(a). i). Identify A and complete the following reactions

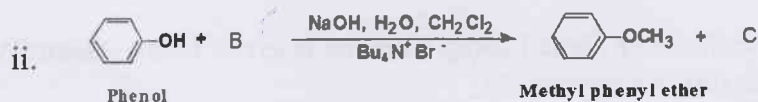
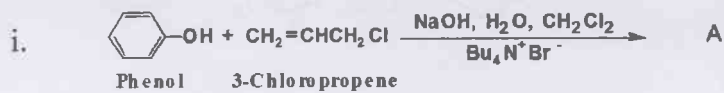


ii). When Styrene treated with methanolic Mercuric trifluoroacetate followed by Sodium borohydrate, a phenolic methyl esters produces. Identify A and B and complete the following reaction



(08 marks)

(b). When Phenol reacts with 3-Chloropropane under given conditions, 'A' produced. However when Phenol reacts with 'B' under same conditions, Methyl phenylether and 'C' produced. Identify A, B and C and complete the following reactions



(06 marks)

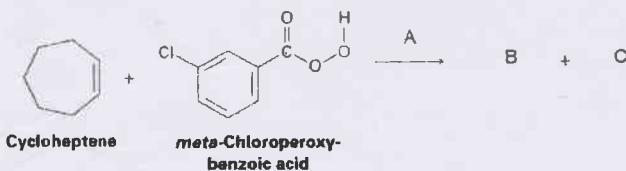
(c). i. Identify X, Y and Z, and complete the following reactions



ii. The Y of the above (i) is an extremely weak base. Explain this by using resonance hybrids

(06 marks)

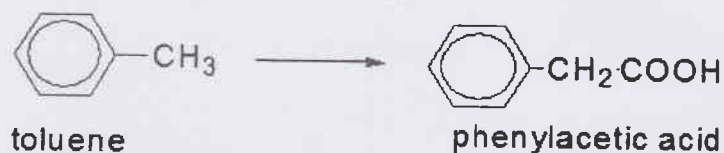
(d). Synthesis of epoxides can be achieved by treatment of an alkene with a peroxyacid. Identify A,B and C and complete the following reaction.



(05 marks)

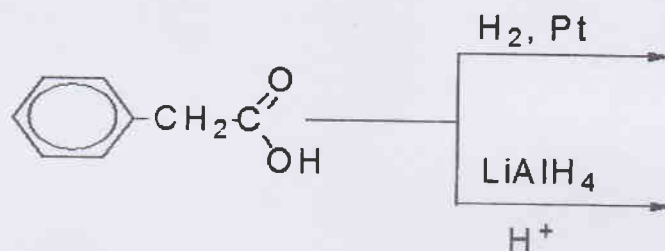
05.

- (a). Write the mechanisms for the synthesis of phenylacetic acid from toluene
- via hydrolysis of a nitrile.
 - by carbonation of Grignard reagent



(10 marks)

- (b). Reduction of phenyl acetic acid using two different conditions are given below. Identify the products and complete the following reactions



(05 marks)

(c).

- List out the type of dyes used in the industry.
- Write a short account on "direct dyes" used to color fabrics

(10 marks)

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